

GUI and Scenario Creation

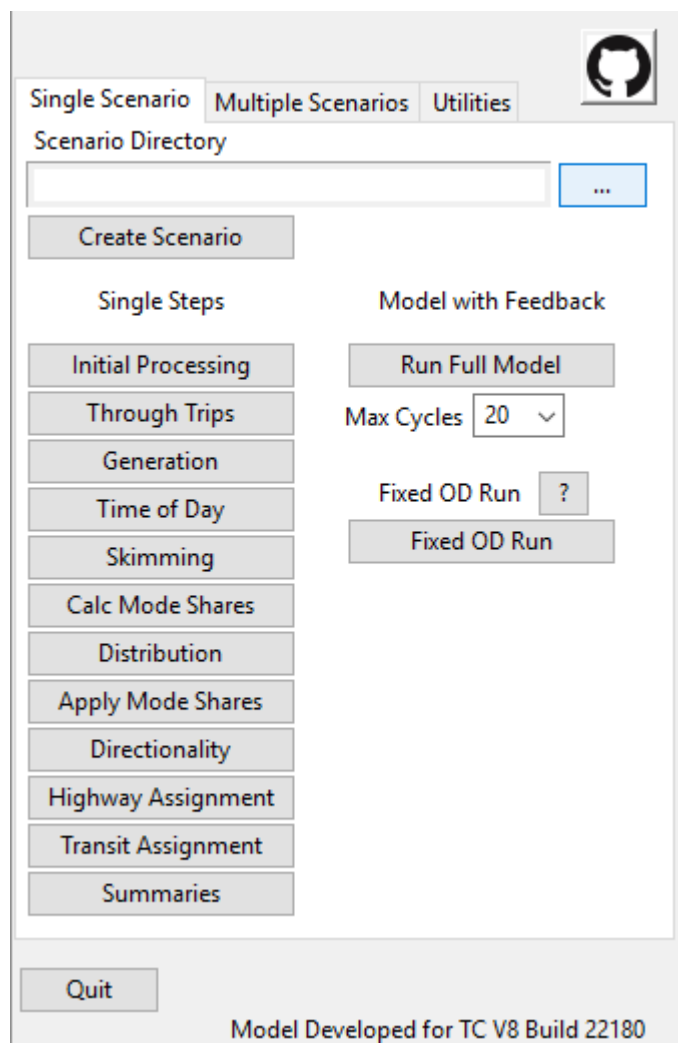
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Kyle Ward edited this page on Jul 9, 2018 · 2 revisions

After [installing](#) the model, you can open the model GUI using:

"Tools" -> "Add-ins" -> "NRV"

You will see the following user interface:



The UI is first organized into three tabs:

- Single Scenario
- Multiple Scenarios
- Utilities

If this is your first time using the UI, it is important that you read this page in its entirety

PRIOR to setting up and running your first scenario. There are many helpful tips and

PRIOR to setting up and running your first scenario, there are many helpful tips and important points that you will likely need to be successful.

Single Scenario

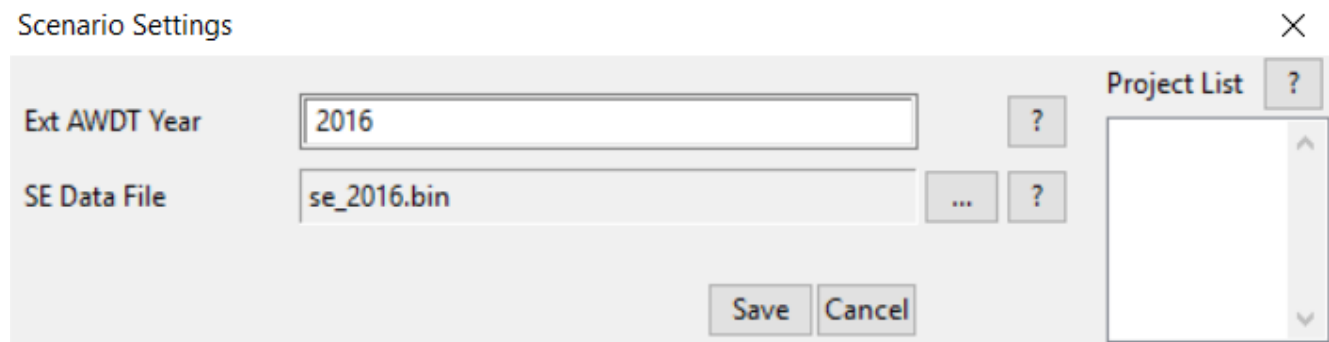
As the name implies, this tab is for running individual scenarios and contains the tools that will be used most often during model application. This tab can be further divided into three areas:

- Create Scenario
- Single Step Buttons
- Full Model Run and Fixed OD Run

Create Scenario



The first step is to select a scenario folder to run. The model comes with the base scenario included. Click the "..." button, select the folder, and press "OK". The "Scenario Settings" screen opens.



First, note the "?" buttons by each setting. These provide additional information as to what each setting means. Also note that the project list is empty, because this scenario is the base year.

The GUI will prevent you from selecting improper settings. For example, it will only allow you to select a year with data available for the external table extrapolation.

Confirm the settings by clicking "Save".

The scenario can now be created by clicking the "Create Scenario" button. This process extracts the scenario input files from the master model directory. Importantly, only input files

extracts the scenario input files from the master model directory. Importantly, only input files are created by this process. Pressing the "Create Scenario" button again will display a warning but will let you overwrite the existing input files. **Any existing output files will not be modified until the model is re-run.**

More detail

This simplified scenario creation operates using "recipe" CSV files stored in the scenario folder.

 HighwayProjectList.csv

 ScenarioSettings.csv

 TransitProjectList.csv

If an empty folder is selected to have a scenario created, these CSV files are created. Otherwise, they are read and updated via the GUI. See the [highway manager](#) page for more details on the HighwayProjectList.csv file. See [transit manager](#) for TransitProjectList.csv

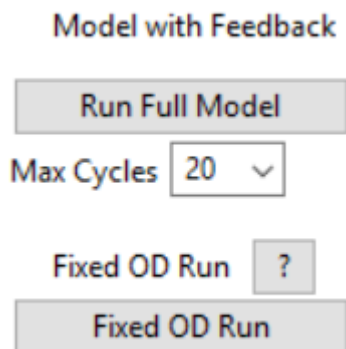
Because scenarios are fully defined by recipes, they can be shared easily between multiple model users. Users can perfectly replicate the scenario on their machine by pasting the recipes into a new scenario folder. This is true even if the model is installed in different directories on different machines. The scenario settings use relative path names.

Single Step Buttons



Each button runs a single step of the model. They are most often used during model development or to debug a scenario that has crashed. **It is important to note that running each of these buttons in order is not the same as a fully-converged model run.**

Full Model Run and Fixed OD Run

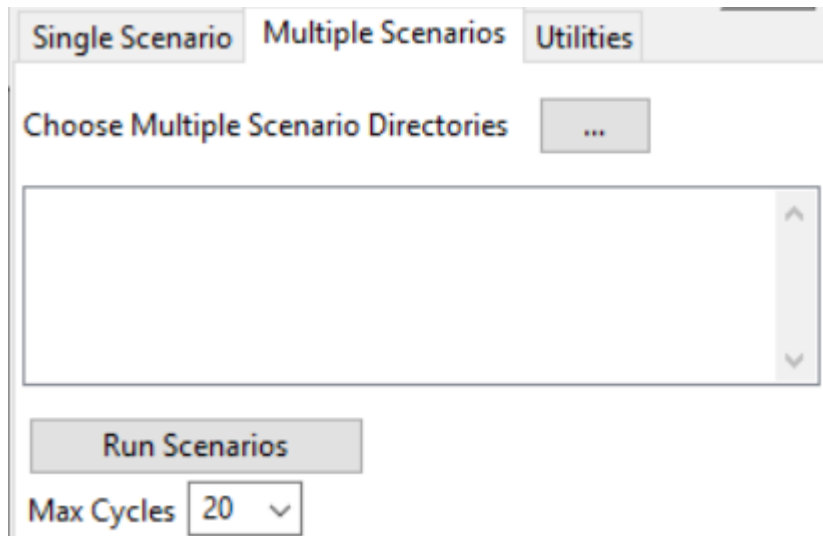


The "Full Model Run" button creates an official model forecast by running model steps through feedback loops to achieve a converged solution. The "Max Cycles" option should generally not be adjusted but can be used to shorten the model run time and produce a rough (unofficial) estimate.

The "Fixed OD Run" supports a common application of travel models: a small change is made to the input highway network and the impact is measured while keeping the trip tables

to the input highway network, and the impact is measured while keeping the trip tables constant from a previous run. Simply copy a previously- run scenario, modify the input highway network and use this option to automatically update highway speeds, capacities, re-run assignment, and update the appropriate summaries.

Multiple Scenarios

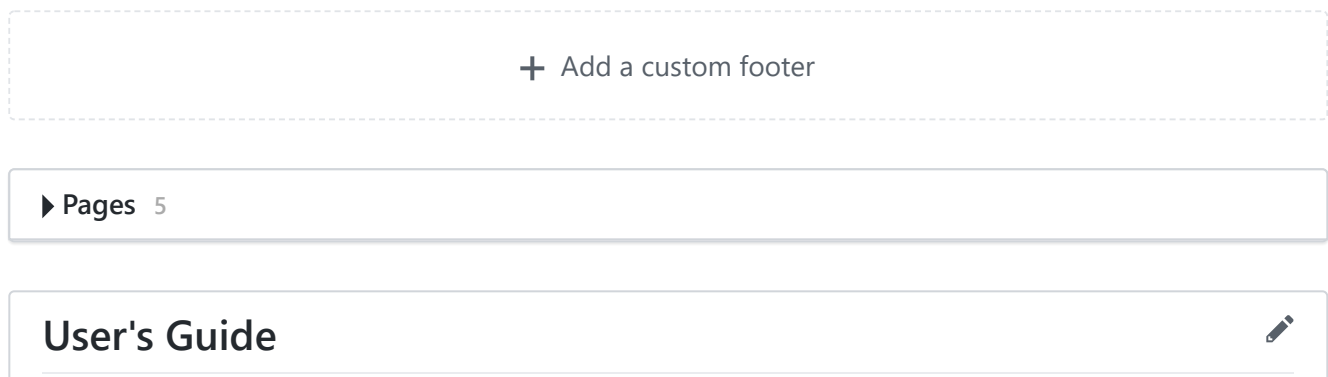


The second GUI tab supports a less-common model application, but one that usually requires a lot of time on the part of the modeler: running multiple scenarios in sequence.

The "..." can be used to select scenario folders one at a time to add to the queue. When each scenario folder is chosen, the GUI automatically checks to see if it has been created. If not, it will prompt you to create it. Scenarios can be removed from the queue by right-clicking and selecting "Remove Scenario". Once all scenarios have been chosen, clicking the "Run Scenarios" button will execute each one in order.

Utilities

This tab provides a space to hold other useful tools (e.g. clearing the work space of all open maps/views and progress bars).



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